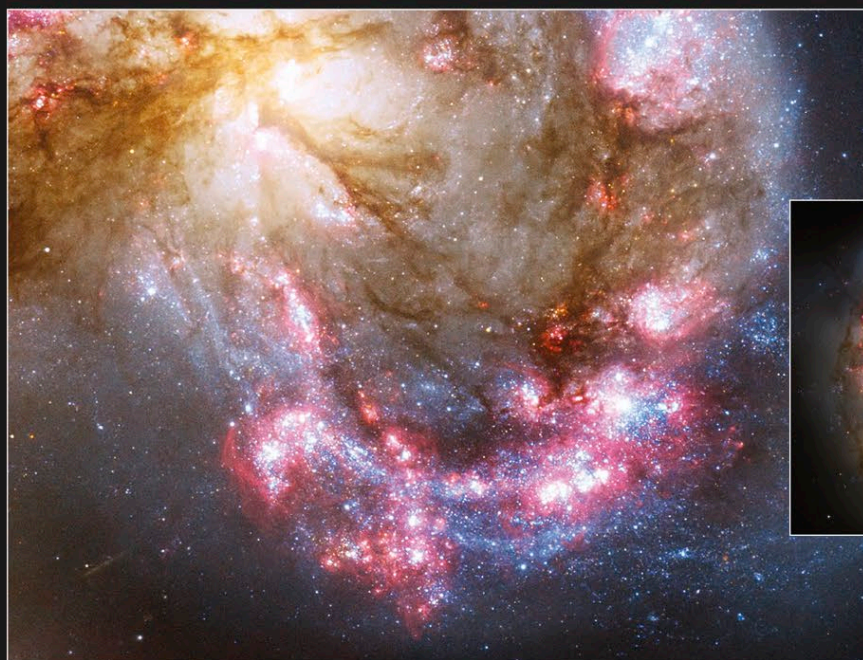
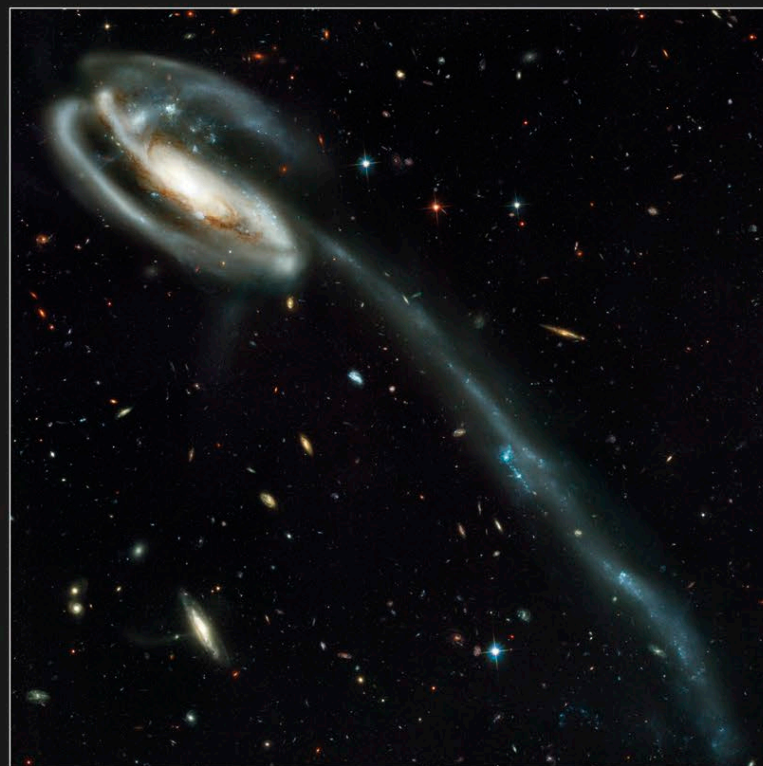


TIDAL FORCES

As two galaxies approach each other, their gravitational fields interact and can affect their shapes. For example, because the galaxies' gravitational fields pull more strongly on the near side of each of the galaxies than on their more distant sides, their near sides become stretched out toward each other. Such gravitational distortion is greater on the less massive of the two galaxies because of the stronger gravity of the other, more massive galaxy. However, the disks of even large spiral galaxies can be warped by the gravity of relatively small neighbors. When spirals collide with one another, one or more of the spiral arms may unwind, transforming into a long trail of stars that stretches out on the opposite side from the collision. Among the best-known examples of these "tidal tails" are the ones associated with the Tadpole Galaxy and the Antennae Galaxies.

THE TADPOLE GALAXY

This galaxy, in the constellation Draco, has a tail that stretches for some 280,000 light-years and is thought to have formed when one of the spiral galaxy's arms unwound in a close encounter with a smaller galaxy.



STARBURSTS

Intergalactic collisions can send immense shock waves through the galaxies involved, compressing large areas of interstellar gas and triggering enormous waves of star formation known as starbursts.

During these events, star birth occurs much faster than normal, giving rise to huge "super star clusters" that may (if they survive) evolve into globular clusters. Starbursts are commonly seen in direct collisions, such as that of the Antennae Galaxies, but can also occur in close encounters between galaxies, as seen in the Cigar Galaxy (see p.314) due to its close encounter with Bode's Galaxy. Radiation from the numerous massive stars being formed, coupled with shock waves from supernovae as the heaviest stars rapidly age and explode, may blow gas and dust out of the galaxy, and it is this dispersion that may ultimately bring the starbursts to an end.

STAR BIRTH IN THE ANTENNAE GALAXIES

This image of the Antennae Galaxies (NGC 4038 and 4039), which lie about 45 million light-years from Earth, reveals stars being born in huge starburst regions. The newborn stars are a brilliant white-blue and are surrounded by glowing pink emission nebulae.